March 19, 2026

Dear Members of the Search Committee,

I am writing to apply for the tenure-track assistant professor at the Toyota Technological Institute at Chicago, with focus on **Computer Vision, Foundation Models, Robotics and Responsible AI**. I am also flexible to discuss possibilities for a Research Assistant Professor if the later is not available. I started as an assistant professor at the University of Ontario Institute of Technology (UOIT), Canada from July'23 and an affiliate assistant professor in the University of British Columbia, Canada since April'24 till May'25. I held multiple grants from NSERC and other funding agencies that amount to more than $300K on learning pixel-level video understanding. Previously, I was a postdoc in York University (VISTA fellowship recipient) with Prof. Richard Wildes and a Vector researcher with Prof. Kostas Derpanis, 2021-2023. In my postdoc, I worked on interpretability, fewshot learning and video understanding. I have obtained my PhD degree in Computing Science from the University of Alberta, Canada, 2015-2021, under the supervision of Prof. Martin Jagersand on the intersection of few-shot learning and video understanding with application to two real-world robotics settings. Through my studies I acquired scholarships amounting to 220K$. Throughout my career, I have had experience working across three robotic platforms; robotic arms, self-driving cars and unmanned aerial vehicles (quadcopters). My current research goals are oriented towards building general-purpose embodied agents within a responsible AI framework. My focus is on embodied AI, video understanding, multi-modal learning with emphasis on interpretability, safety and data efficient learning for a responsible AI approach. Additionally, I enjoy being an educator to motivate students to learn more about the basics of mathematics, machine learning, computer vision, robotics and responsible AI. I believe that my previous research and teaching experience makes me a strong candidate for this position. I am a female African, which is an under-represented group in diversity, and I am also a Canadian citizen.

My main research contribution is establishing the topic that investigates few-shot learning and video segmentation and devising multiple techniques to benefit from the temporal structure in the data within a data efficient learning framework for robotics applications. When I started most of the few-shot learning research was limited to simple image classification at that time. Additionally, I innovated an approach benefiting from graph theory and spectral clustering towards an end-to-end spatiotemporal label propagation in video transformers, and mentored a PhD student in its implementation. I have also contributed in extending the aforementioned to robotics, specifically robot manipulation and autonomous driving. I have devised the first deep class agnostic motion segmentation model for autonomous driving in collaboration with Valeo Vision Systems within multi-task learning systems towards a safety critical approach. Additionally, I have participated in the KUKA innovation competition part of team Alberta, where we received a finalist award. We explored online tool and task learning using human robot interaction [1]. Currently, I am working on adapting multi-modal large language models to pixel-level understanding specifically segmentation, where I have developed the PixFoundation Series[2], and exploring their adaptation to embodied AI. I am also working on building a universal dense prediction tasks data engine that relies on widely adopted foundation models and forms of transduction, with focus on video segmentation and depth. I aim to use the data from the aforementioned to train a 4D motion-centric video transformer for dense prediction that utilizes multi-images in videos unlike single-image models. I regularly publish in top tier venues in computer vision **(CVPR, ICCV, TPAMI, IJCV, WACV)**, robotics **(ICRA, IROS, IJCAI)** and transportation systems (ITSC, IV). Some of these publications are on data-efficient learning, some are on interpretability, and others are on AI safety, where all are at the core of responsible AI. The majority of these works I am either the leading first author or equally contributing or a senior last author. In addition to my publications that are on purely responsible AI topics discussing inequity in the computer vision field with focus on African researchers; **(EAAMO, JAIR)**.

I supervised one research scientist in UBC, served on the supervisory committee of one PhD student in UBC who graduated last year and graduated two Master's level students in AMMI/AIMS Senegal and UOIT. Furthermore, I hosted one visiting PhD student in Summer 2023 and 2024. Through my supervision, I was able to guide the PhD student in UBC on top venues publications in CVPR 2024 and TMLR 2026 focusing on multimodal learning, data efficient learning and large language models. I also guided the visiting PhD student to work on the topic of responsible AI with focus on African Computer Vision research. Moreover, we worked together on the intersection of video understanding and computational neuroscience, which resulted in a Nature Scientific Reports publication. Additionally, I supervised the MSc student in AMMI on studying multi-modal large language models for remote sensing within a vision centric evaluation in collaboration with the research scientist in my lab. I also graduated one Master's level student working on developing multiscale video transformers for autonomous driving. During my postdoc, I mentored and guided a PhD student to innovate unified multiscale spatiotemporal modeling in video transformers. I have also mentored another four PhD students in both York University and UBC on their thesis research and publications. These include works on interpretability, video understanding, tracking and ego-centric vision.

I am mainly interested in that position as I am quite passionate about both research and teaching. My research agenda aligns well with TTIC and their faculty. I see it as a good opportunity to build a community of students that can learn more about computer vision, robotics, responsible AI and deep learning. I have good experience teaching the theoretical concepts on probabilistic machine learning in Winter 2024 in UOIT, Canada for undergraduate students. Moreover, I have experience teaching classical computer vision and 3D vision as a co-instructor in Winter 2021 in the University of Alberta, Canada in the Multimedia program for MSc students. I have good experience teaching Discrete Mathematics (Fall 2023 & 2024) and Data Structures (Winter 2024) in UOIT, Canada. I also have experience teaching Computer Vision tutorials in preparation for the African Computer Vision Summer School as a supporting organizer, remotely [3].

I actively work and publish in top venues in both Computer Vision and Robotics, a combination that is

[1] `https://www.youtube.com/watch?v=aLcw73dt_Oo`
[2] `https://msiam.github.io/PixFoundationSeries/`
[3] `https://youtu.be/6_WbhllqKRk?t=5095`

rare among female African researchers. I have consistently contributed to the Computer Vision field not only in research but through my service. I organized for the past four years workshops in CVPR on a yearly basis including the latest PixFoundation Workshop [4]. I have also been an AC in WACV for the past 3 years, 2024-2026 and a Senior-PC in IJCAI26. I also founded Ro'ya - Computer Vision for Africa[5] community that is meant specifically to empower Africans in the field, which resulted in various events and publications. In summary, I believe that my research, teaching, mentoring/supervising students, and my background can prove to be a good addition to your department. Thank you for your consideration and I am looking forward to your response.

Sincerely,
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[4] `https://sites.google.com/view/pixfoundation`
[5] `https://x.com/RoyaCV4Africa/status/1807054249777332632`